

Molecular mimicry in the autoimmune pathogenesis of rheumatic heart disease.

Molecular mimicry is a hallmark of the pathogenesis of rheumatic fever where the streptococcal group A carbohydrate epitope, N-acetyl glucosamine, and the α -helical coiled-coil streptococcal M protein structurally mimic cardiac myosin in the human disease, rheumatic carditis, and in animal models immunized with streptococcal M protein and cardiac myosin. Recent studies have unraveled the potential pathogenic mechanisms by which the immune response against the group A streptococcus attacks the rheumatic valve leading to chronic rheumatic heart disease. Both B- and T-cell responses are involved in the process, and evidence for the hypotheses of molecular mimicry and epitope spreading are reviewed.